

Task-1

1. Name 5 IT job roles and write one line about what each role does.

1. **Software Developer** – Designs, writes, and maintains software applications.
 2. **Web Developer** – Builds and maintains websites and web applications.
 3. **Mobile App Developer** – Creates applications for Android and iOS devices.
 4. **Software Tester (QA Engineer)** – Tests software to find bugs and ensure quality.
 5. **Database Administrator (DBA)** – Manages databases and keeps data secure and organized.
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2. List 3 product companies and 3 service companies with one example of work each does.

Product Companies

1. **Microsoft** – Develops Windows operating system and Microsoft Office.
2. **Google** – Develops Android OS and Google Search.
3. **Adobe** – Develops Photoshop and Acrobat software.

Service Companies

1. **TCS (Tata Consultancy Services)** – Provides software development and IT consulting.
 2. **Infosys** – Offers application development and digital transformation services.
 3. **Wipro** – Provides cloud computing, cybersecurity, and IT support services.
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3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Term	Full Form	Use
GitHub	Git + Hub (Git-based code hosting platform)	Stores code and helps developers collaborate using Git.
Slack	Searchable Log of All Conversation and Knowledge	Used for team communication and collaboration.
Jira	(Officially a product name, not an acronym)	Used to track bugs, tasks, and software projects.
IDE	Integrated Development Environment	Software used to write, run, and debug code.
API	Application Programming Interface	Allows different applications to communicate with each other.

4. Name any 5 programming languages and mention one popular use case for each.

Programming Language	Popular Use Case
Python	Artificial Intelligence and Data Science
Java	Android application development
C++	Game development
JavaScript	Interactive websites
Dart	Flutter mobile app development

5. What is the difference between a developer and a tester? (2 lines)

- A **Developer** writes and builds software applications by creating code.
- A **Tester (QA Engineer)** checks the software for bugs and ensures it works correctly before release.

Task-2

1. What are the three main layers of software architecture? Write one line about each.

1. **Presentation Layer (UI Layer)** – Displays information to users and accepts user input.
 2. **Business Logic Layer (Application Layer)** – Processes data and implements the application's rules and logic.
 3. **Data Layer (Database Layer)** – Stores, retrieves, and manages application data.
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2. Define client and server in your own words with one real-life example.

- **Client:** A client is a device or application that requests data or services from a server.
- **Server:** A server is a computer or system that provides data or services to clients.

Real-life Example:

When you open **YouTube** on your phone, your phone is the **client**, and YouTube's computers that send videos to your phone are the **servers**.

3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Desktop Apps	Web Apps	Mobile Apps
Installed on a computer.	Runs in a web browser.	Installed on a smartphone or tablet.
Can work without the internet (depending on the app).	Usually requires an internet connection.	Can use phone features like camera, GPS, and notifications.

4. Name any 3 databases and mention whether each is SQL or NoSQL.

Database	Type
MySQL	SQL
PostgreSQL	SQL
MongoDB	NoSQL

Task-3

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Term	Full Form	Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	A secure version of HTTP that encrypts data during communication.
FTP	File Transfer Protocol	Used to upload and download files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used to send emails from one mail server to another.

2. What is the difference between IPv4 and IPv6? (2 lines)

- **IPv4** uses **32-bit addresses** (e.g., 192.168.1.1) and supports about **4.3 billion** unique addresses.
 - **IPv6** uses **128-bit addresses** (e.g., 2001:db8::1) and supports a much larger number of unique addresses with improved security and efficiency.
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3. Run `nslookup google.com` on your system and write down the IP address you get.

I can't run commands on your computer, so I can't see your result.

Open **Command Prompt** and run:

```
nslookup google.com
```

Example output (your result may be different):

```
Name:      google.com
Address: 142.250.183.14
```

Note: The IP address may vary depending on your location and Google's servers.

4. What do these HTTP status codes mean?

Status Code	Meaning
200 OK	The request was successful.
301 Moved Permanently	The requested page has been permanently moved to a new URL.
403 Forbidden	The server understood the request but denies access.
404 Not Found	The requested page or resource was not found.
500 Internal Server Error	The server encountered an unexpected error while processing the request.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

DNS Level	Example	Explanation
Root	.	The highest level in the DNS hierarchy (usually hidden).
TLD (Top-Level Domain)	.com	The top-level domain that identifies the domain category.
Domain	google	The registered name of the website.
Subdomain	www	A subdivision of the main domain, commonly used for websites.

Example Breakdown:

www.google.com.

```
graph TD
    Root[.] --- TLD[.com]
    TLD --- Domain[google]
    Domain --- Subdomain[www]
```

TLD (.com)
Domain (google)
Subdomain (www)

Task-4

1. List all 6 phases of SDLC in order with one line about each.

1. **Requirement Analysis** – Gather and understand the customer's requirements.
 2. **System Design** – Create the architecture and design of the software.
 3. **Development (Coding)** – Write the source code based on the design.
 4. **Testing** – Test the software to find and fix bugs.
 5. **Deployment** – Release the software to users for use.
 6. **Maintenance** – Update, improve, and fix issues after deployment.
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2. Write 3 differences between Agile and Waterfall models.

Agile	Waterfall
Development is done in small iterations (Sprints).	Development is completed in one sequential process.
Requirements can change during the project.	Requirements are fixed before development begins.
Customer feedback is taken regularly.	Customer feedback is usually taken after the project is completed.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

1. **Product Owner** – Defines requirements and prioritizes the product backlog.
 2. **Scrum Master** – Helps the team follow Scrum practices and removes obstacles.
 3. **Development Team** – Designs, develops, tests, and delivers the product.
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4. What is a Sprint in Scrum? Mention its typical duration.

A **Sprint** is a fixed period during which the Scrum team completes a set of planned tasks and delivers a working product increment.

Typical Duration: 1 to 4 weeks (most commonly **2 weeks**).

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Jira Column	Meaning
Backlog	A list of tasks that are planned but not yet started.
To Do	Tasks that are ready to be started.
In Progress	Tasks that are currently being worked on.
Done	Tasks that have been completed successfully.

Task-5

1. Write the Git command for each.

Task	Git Command
Initialize a repository	<code>git init</code>
Stage all files	<code>git add .</code>
Commit with a message	<code>git commit -m "Initial commit"</code>
Push to remote	<code>git push origin main</code>
Pull latest changes	<code>git pull origin main</code>

2. What is the difference between `git fetch` and `git pull`? (2 lines)

- `git fetch` downloads the latest changes from the remote repository but does **not** merge them into your current branch.
 - `git pull` downloads the latest changes **and automatically merges** them into your current branch.
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3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

I can't create a GitHub repository or push files from your computer.

After creating the repository and pushing your file, share the repository URL in this format:

`https://github.com/your-username/repository-name`

Example:

`https://github.com/johndoe/git-practice`

4. What is the difference between a Fork and a Pull Request? (2 lines)

- A **Fork** creates your own copy of someone else's GitHub repository in your account.
 - A **Pull Request (PR)** is a request to merge your changes into another repository or branch.
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5. Create a new branch named `feature-test`, make one commit on it, and merge it back to the main branch. Write the commands you used.

```
# Create and switch to a new branch
git checkout -b feature-test

# Create or edit a file
echo "Feature added" > feature.txt

# Stage the file
git add feature.txt

# Commit the changes
git commit -m "Add feature.txt"

# Switch back to the main branch
git checkout main

# Merge the feature-test branch
git merge feature-test

# (Optional) Delete the merged branch
git branch -d feature-test
```